



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Pizza Store Analysis

By

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About me

Sanskruti Ravindra Patil

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Why I Want to Learn Data Science

I want to learn data science because it empowers me to turn raw information into meaningful insights.

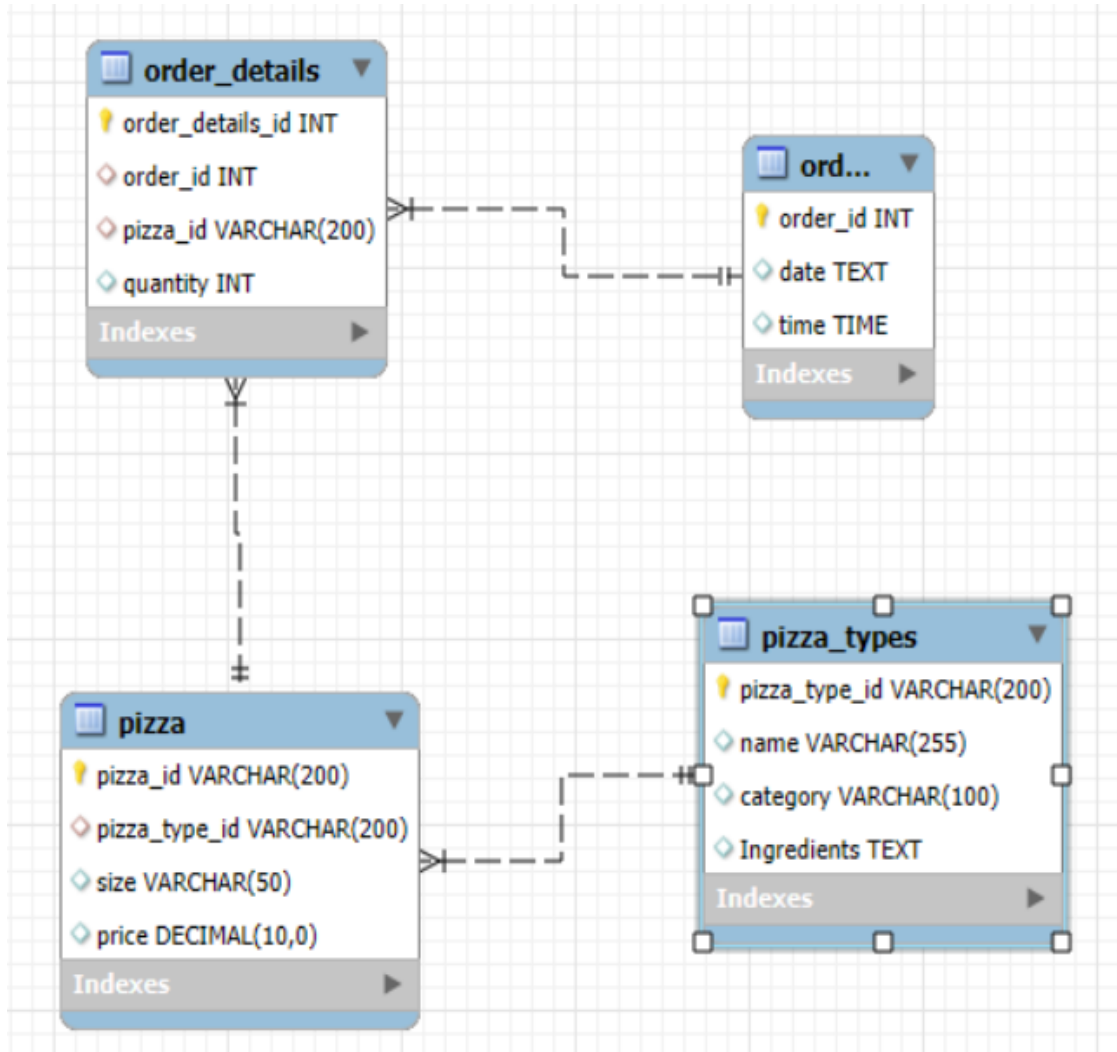
Profiles:

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- GitHub: <https://lnkd.in/d65hKDjZ>

Objective of the Project :

- Design a relational database to model pizza sales, including pizza types, sizes, orders, and ingredients.
- Implement normalized schema using SQL with appropriate primary and foreign key constraints to ensure data integrity.
- Perform multi-level analysis using SQL queries to extract insights across basic, intermediate, and advanced levels.
- Analyze customer behavior through time-based order patterns and category wise preferences.
- Evaluate product performance by ranking pizzas based on quantity sold and revenue contribution.

ER Diagram and Schema :



- **Orders table:** It gives information like order ID, date, and time. Each order can include multiple pizzas, which are recorded in the Order_Details table.
- **Order Details table:** It acts as a link between Orders and Pizzas, it stores which pizzas were ordered and in what quantity.
- **Pizza table:** It contains details such as pizza ID, size, and price. Each pizza also belongs to a specific type, which is defined in the Pizza_Types table.
- **Pizza_Types table:** It stores information like the pizza name, category, and ingredients.

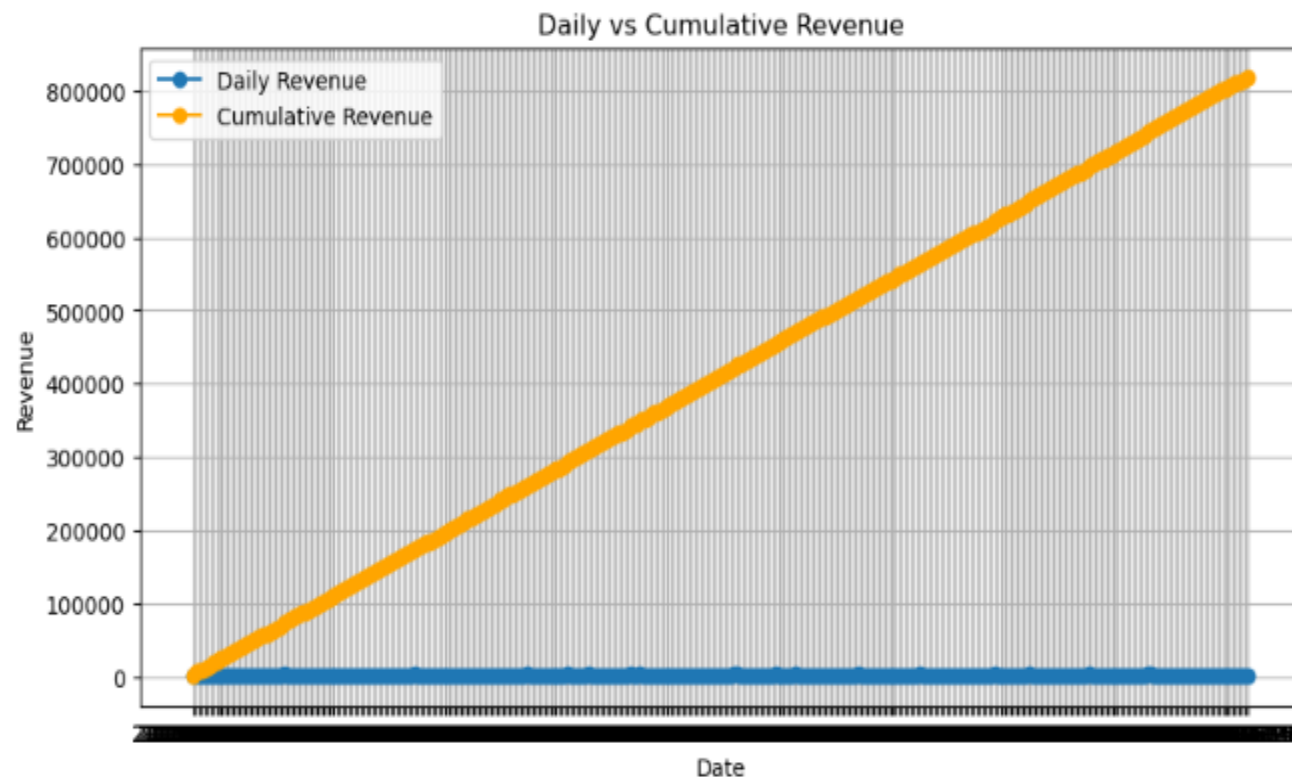
Key Areas of Exploration :

- Daily and Cumulative Revenue.
- List the top 5 most ordered pizza types along with their quantities.
- Identify the most common pizza size ordered.
- Find the category-wise distribution of pizzas (count of pizza types per category).
- Determine the distribution of orders by hour of the day.

Daily and Cumulative Revenue.

```
SELECT o.date,  
       SUM(od.quantity * p.price) AS daily_revenue,  
       SUM(SUM(od.quantity * p.price)) OVER (ORDER BY o.date) AS cumulative_revenue  
FROM orders o  
JOIN order_details od ON o.order_id = od.order_id  
JOIN pizzas p ON od.pizza_id = p.pizza_id  
GROUP BY o.date  
ORDER BY o.date;
```

	date	daily_revenue	cumulative_revenue
▶	2015-01-01	2713.85000000000004	2713.85000000000004
	2015-01-02	2731.89999999999996	5445.75
	2015-01-03	2662.4	8108.15
	2015-01-04	1755.45000000000003	9863.6
	2015-01-05	2065.95	11929.55
	2015-01-06	2428.95	14358.5
	2015-01-07	2202.20000000000003	16560.7
	2015-01-08	2838.34999999999995	19399.05
	2015-01-09	2127.35000000000004	21526.4
	2015-01-10	2463.95	23990.3500000000002
	2015-01-11	1872.30000000000002	25862.65
	2015-01-12	1919.05000000000002	27781.7
	2015-01-13	2049.60000000000004	29831.3000000000003
	2015-01-14	2527.39999999999996	32358.7000000000004
	2015-01-15	1984.80000000000002	34343.5000000000001
	2015-01-16	2594.15	36937.6500000000001
	2015-01-17	2064.10000000000004	39001.7500000000001
	2015-01-18	1976.85000000000001	40978.6000000000006
	2015-01-19	2387.14999999999996	43365.7500000000001

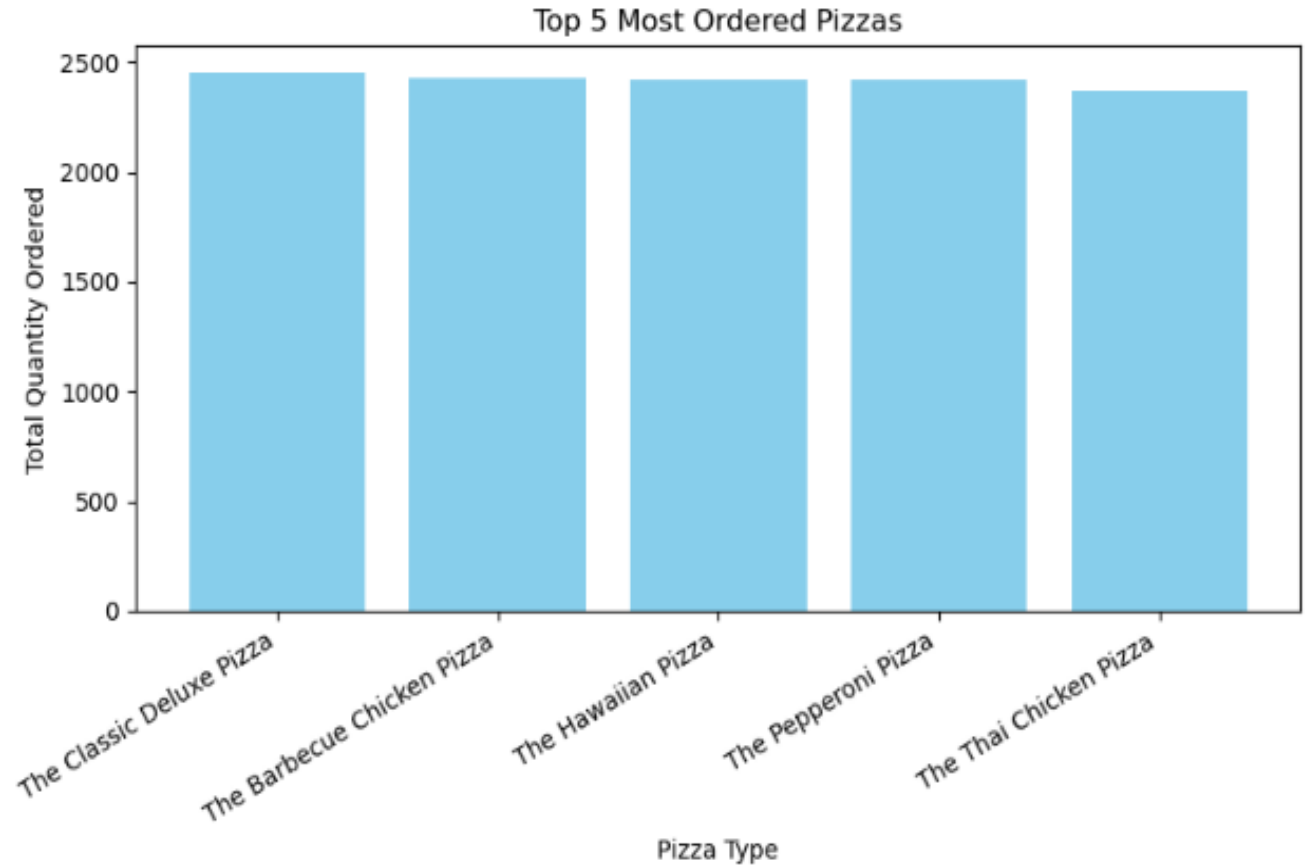


- ❑ The blue line shows the Daily Revenue — money earned from pizza sales each day.
- ❑ The orange line shows the cumulative revenue, which keeps adding up the daily revenues to show the total revenue growth over time.
- ❑ This visualization helps track both short-term performance (daily sales) and long-term progress (total revenue).

List the top 5 most ordered pizza types along with their quantities.

```
SELECT pt.name, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Result Grid	Filter Rows:
name	total_quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371



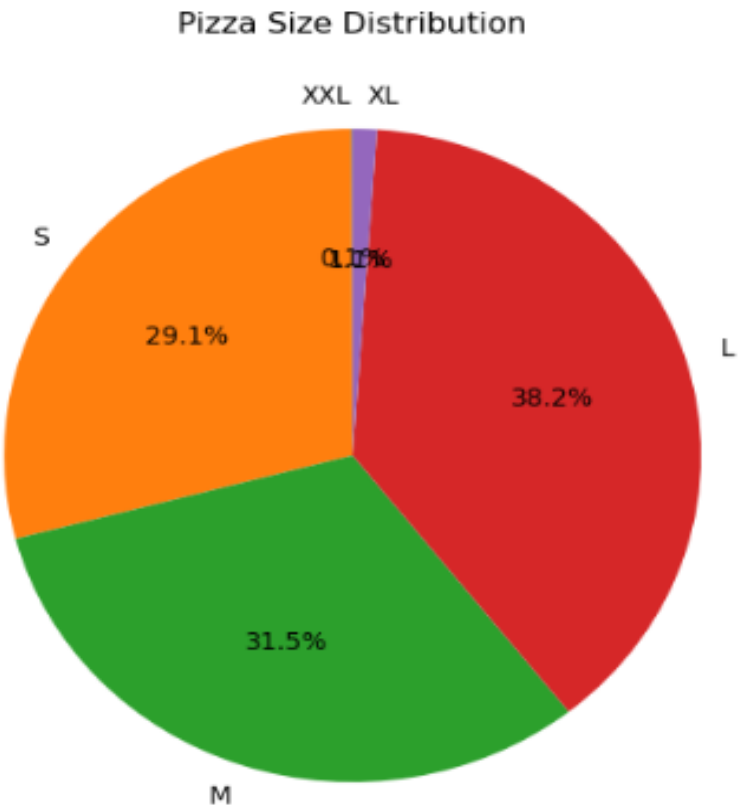
- ❑ It combines data from the order details, pizza, and pizza type tables to calculate the total quantity sold for each pizza.
- ❑ The results are sorted in descending order to find the most popular pizzas. The bar chart visually highlights which pizzas customers prefer.
- ❑ The most — useful for sales insights and marketing strategies.”

Identify the most common pizza size ordered.

```
SELECT p.size, SUM(od.quantity) AS total_quantity
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY FIELD(p.size, 'S','M','L','XL');
```

Result Grid			Filter Rows:
	size	total_quantity	
▶	XXL	28	
	S	14403	
	M	15635	
	L	18956	
	XL	552	

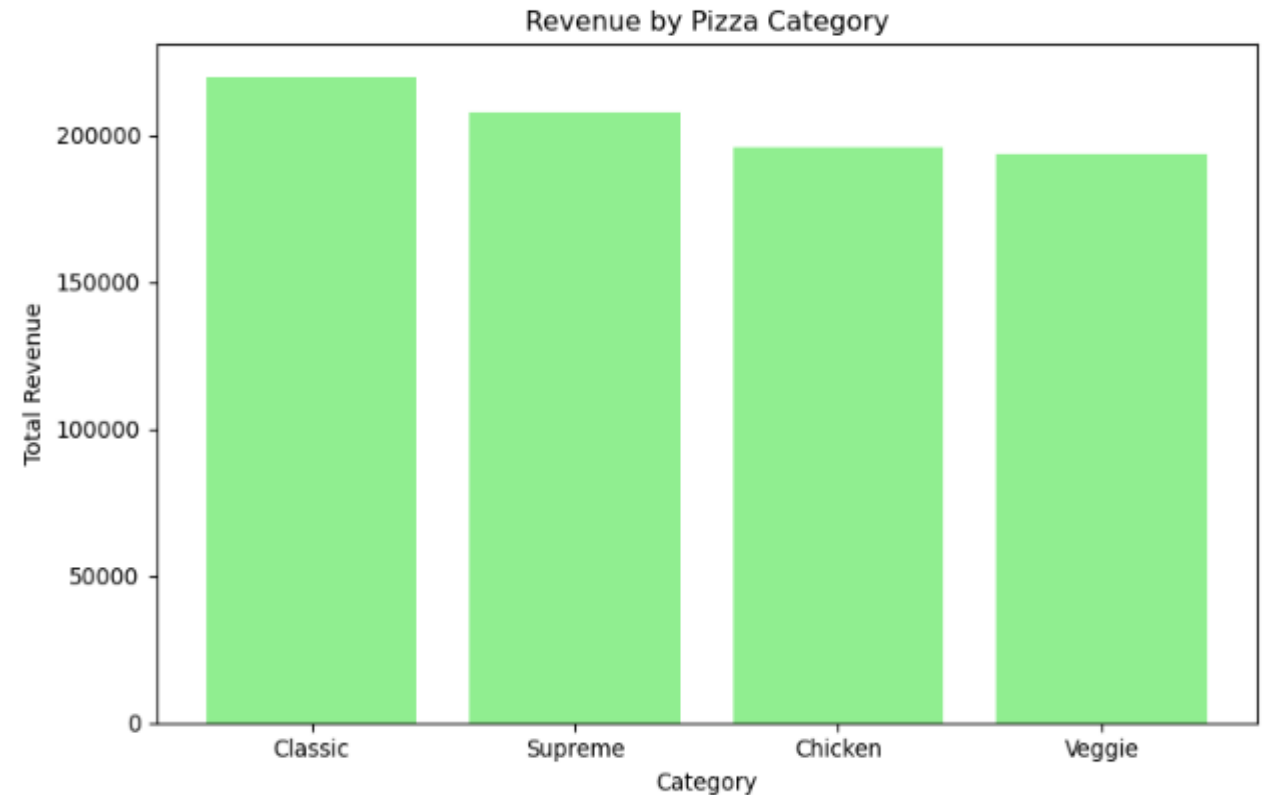
- ☐ Pie Chart: Shows the percentage share of each pizza size in total orders.
- ☐ Reveals which size is most popular among customers.
- ☐ Helps with menu planning, stock management, and customer preference analysis.



Find the category-wise distribution of pizzas (count of pizza types per category).

```
SELECT category, COUNT(*) AS pizza_type_count
FROM pizza_types
GROUP BY category
ORDER BY pizza_type_count DESC;
```

Result Grid	Filter Rows:
category	pizza_type_count
Supreme	9
Veggie	9
Classic	8
Chicken	6

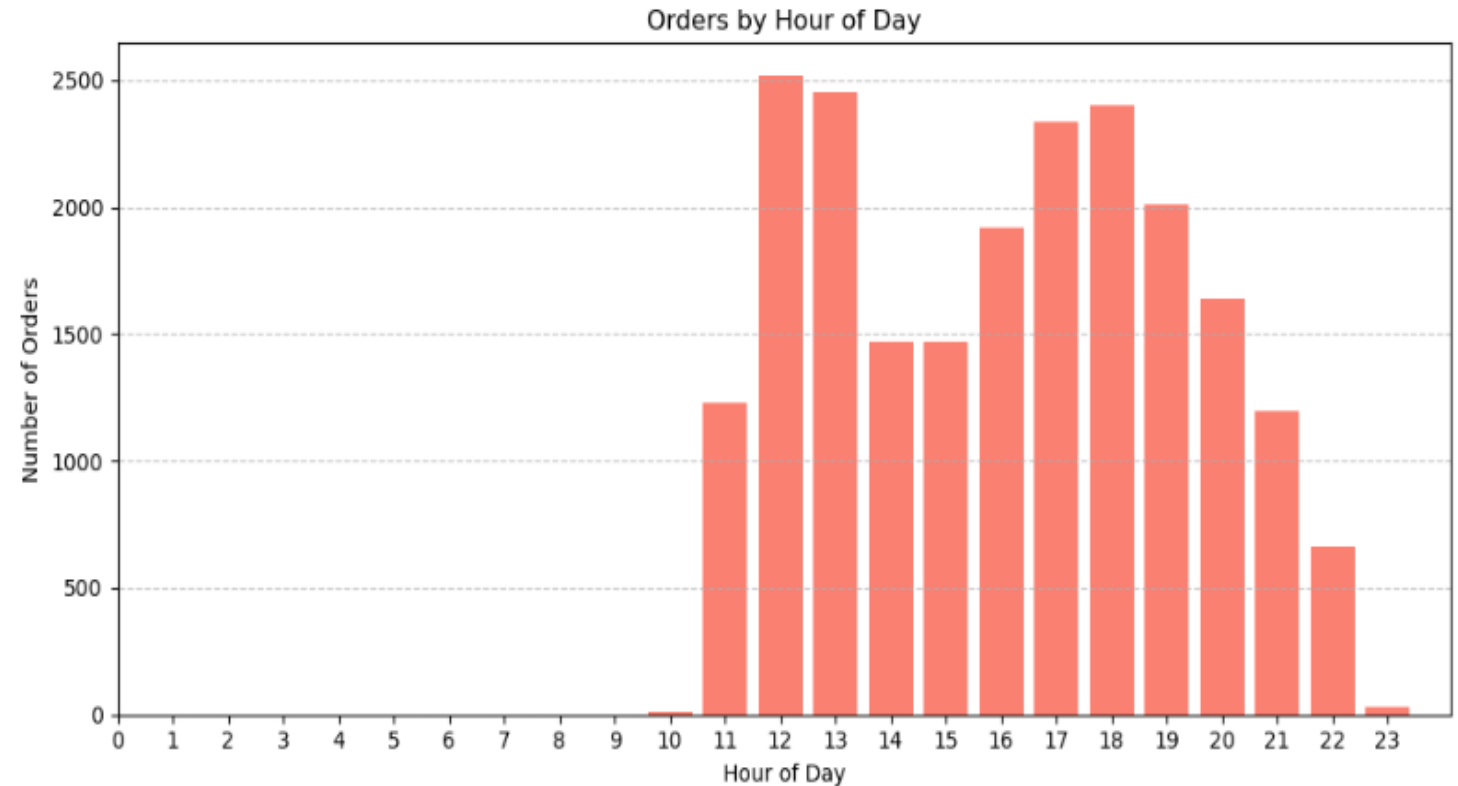


- ❑ Bar Chart: Displays total revenue per category.
- ❑ Highlights which pizza categories generate the most income.
- ❑ Helps businesses focus on best-selling categories and optimize pricing or promotions.

Determine the distribution of orders by hour of the day.

```
SELECT HOUR(time) AS order_hour, COUNT(*) AS total_orders
FROM orders
GROUP BY order_hour
ORDER BY order_hour;
```

Result Grid	Filter Rows:
order_hour	total_orders
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28



- ☐ The bar chart shows the number of orders placed each hour of the day.
- ☐ Helps identify peak order times and customer activity patterns.
- ☐ Useful for staff scheduling, delivery planning, and marketing timing.

Conclusion :

- ❑ Shows steady revenue growth and clear sales trends.
- ❑ The top-selling pizzas The Thai Chicken pizza.
- ❑ The most preferred pizzas size is large.
- ❑ Reveals customer behavior patterns by category and time.
- ❑ Marketing and promotions, Menu optimization, Inventory management, Operational efficiency.

THANK
YOU

